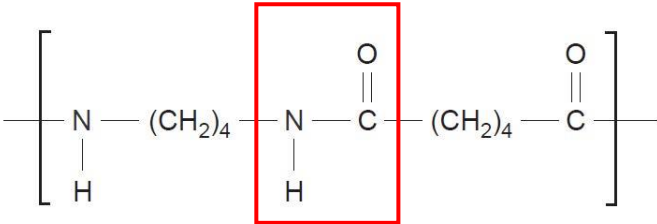
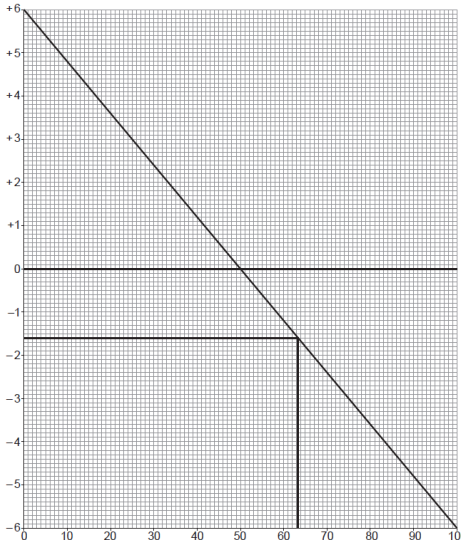


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		award (1) for correct product NH ₃ award (1) for correct balancing (only if product is correct) 2 NaOH 2 NH ₃		2		2		
		(ii)		$0.075 \times \frac{60}{100} = 0.045$ mol of the diamine formed (1) mass of diamine = $0.045 \times 88 = 3.96$ g (1)	1	1		2	1	
		(iii)			1			1		
		(iv)	I	award (1) for any of following only one stage involved therefore quicker only one stage involved therefore probably higher percentage yield fewer waste products therefore greener			1	1		
			II	award (1) for either of following no more effervescence / bubbles (of nitrogen) evolved no more precipitation of the acid			1	1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		elimination of water	1			1		
		(ii)		$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{COOH} \\ \\ \text{CH}_2\text{C}_6\text{H}_5 \end{array}$	1			1		
		(iii)		distance travelled by spot = $8.0 \times 0.54 = 4.3 \text{ cm}$ (1) spot marked on diagram at 5.3 cm (1)	1	1		2	1	
	(c)			 appropriate scale (1) lines drawn correctly (1) award (1) for correct values (+) 36-37% (-) 63-64%			3	3	2	
				Question 11 total	5	4	5	14	4	1